

MA378 Chapter 3: Numerical Integration

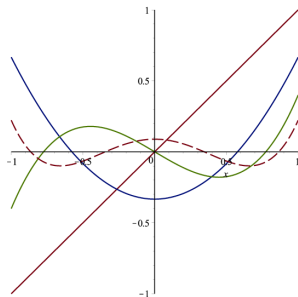
§3.5 Orthogonal Polynomials

Dr Niall Madden

(Niall.Madden@UniversityOfGalway.ie)

Start: 6 March 2026 (W08.2)

And Finished: 13 March (W09.2)



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5.0 The span of topics for today:

- 1 Orthogonal Polynomials
- 2 Inner products
 - Properties
- 3 Sequence of monic polynomials
- 4 Constructing the Sequence
- 5 Properties of the sequence
- 6 Legendre Polynomials
- 7 Exercises

5.1 Orthogonal Polynomials

High order Newton-Cotes methods are of little use because of the problems associated with interpolation by high degree polynomials at equally spaced points. However, high-order Gaussian methods are very useful.

Deriving such methods by undetermined coefficients is not practical, however. There is a simpler way, but some mathematical preliminaries are required, including the ideas of **vector spaces** and **inner products**.

Definition 5.1 (Vector space: informal)

A **vector** space is a collection of objects, called **vectors**, where it makes sense to

- ▶ add two vectors to get another one;
- ▶ multiply a vector by a scalar to get another one.

5.2 Inner products

Definition 5.2 (Vector Space: formal)

V is a *vector space* (a.k.a., a *linear space*) over a field F (e.g, the real or complex numbers) if for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in F$:

- (i) $\mathbf{u} + \mathbf{v} \in V$ (closed under addition)
- (ii) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (Commutativity)
- (iii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (Associativity)
- (iv) V has a zero vector $\mathbf{0}$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
- (v) $-\mathbf{u} \in V$
- (vi) $a\mathbf{u} \in V$
- (vii) $a(b\mathbf{u}) = (ab)\mathbf{u}$
- (viii) F contains 0 and 1 such that $1\mathbf{u} = \mathbf{u}$, $0\mathbf{u} = \mathbf{0}$.
- (ix) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$, and $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$.

5.2 Inner products

Examples:

1. Vectors in $\mathbb{R}^3$

$$u = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad w = \begin{pmatrix} -\pi \\ e \\ 12 \end{pmatrix}$$

2. The vector space, $\mathbb{P}_n$, of polynomials of degree n .

Eg $\mathbb{P}_2$.

$$u = x^2$$

$$v = 3x^2 - 5.1$$

$$w = x^2 - x - 1$$

$$z = 3x + 12.5$$

For us, here are the most important facts about (real) vector spaces:

Linear combination: Given any set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\} \in V$ we call $\mathbf{w} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m$ a **linear combination**. Note: $\mathbf{w} \in V$.

Spanning Set: The set of all possible linear combinations of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is their **spanning set**.

Linear Independence: A set of vectors is **linearly independent** in no element is a linear combination of the others.

 **Basis:** If the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ spans all of V , and its linearly independent, then it is called a **Basis** of V . 

Dimension: **FACT:** a vector space, V , will have many bases, but they all have the same number of elements. That number is called the **dimension** of V .

Definition 5.3 (Inner Product)

Let V be a **real** vector space. An **Inner Product** (IP) is a **real-valued** function $(\cdot, \cdot)$ on $V \times V$ such that, for all $f, g, h \in V$,

- (i) $(f + g, h) = (f, h) + (g, h)$,
 - (ii) $(\lambda f, g) = \lambda(f, g)$, for $\lambda \in \mathbb{R}$.
 - (iii) $(f, g) = (g, f)$,
 - (iv) $(f, f) \geq 0$. $(f, f) = 0 \Leftrightarrow f \equiv 0$.
- $\left. \begin{array}{l} \text{(i)} \\ \text{(ii)} \end{array} \right\} \text{Linearity}$
 $\text{(iii)} \rightarrow \text{Symmetry}$
 $\text{(iv)} \rightarrow \text{"positivity!"}$

Example 5.4

Let $\mathbb{R}^n$ be our vector space, with $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)^T$. Then the following is an inner product:

$$(\mathbf{x}, \mathbf{y}) := \sum_{i=1}^n x_i y_i,$$

eg $n=2$.

$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\mathbf{y} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$(\mathbf{x}, \mathbf{x}) = x_1^2 + x_2^2 = 2.$$

$$(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 = 0$$

Example 5.5

The set of real-valued functions that are continuous and defined on the interval $[a, b]$, denoted $C[a, b]$, is a vector space. And

$$(f, g) := \int_a^b f(x)g(x)dx, \quad (1)$$

is an inner product.

Are we sure this is an IP? Check!

$$(i) \quad (f+g, h) = \int_a^b (f+g)h \, dx = \int_a^b fh \, dx + \int_a^b gh \, dx \\ = (f, h) + (g, h) \quad \checkmark$$

$$(iv) \quad (f, f) = \int_a^b (f(x))^2 \, dx \geq 0. \quad \left| \begin{array}{l} \text{And} \\ (f, f) = 0 \\ \Leftrightarrow f = 0 \end{array} \right.$$

5.3 Sequence of monic polynomials

(See Lecture 23 of Stewart's "Afternotes" for more details).

Definition 5.6 (Monic Polynomial)

A polynomial is *monic* if the coefficient of its leading term is 1.

Examples: $p(x) = 1$ is monic

$$p(x) = x + 2, \quad q(x) = 3x^2 - x^5 + x^9 + 34$$

are monic. So are $\{1, x, x^2, x^3, \dots\}$

Not monic:

$$p(x) = 2$$

$$q(x) = x + 2 + 3x^2$$

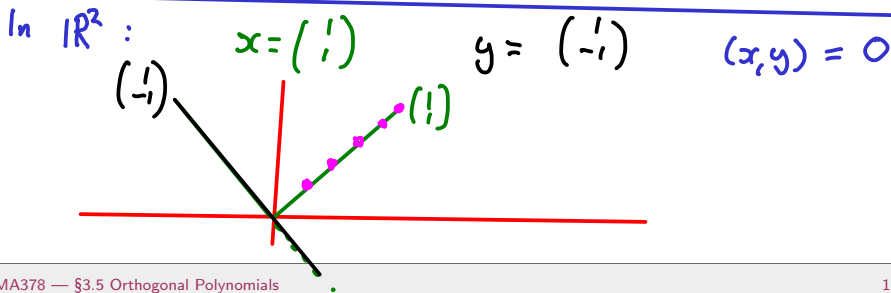
$$r(x) = 3x^2 - x^5 - x^9 + 34$$

5.3 Sequence of monic polynomials

Definition 5.7

Two elements a, b , of a vector space are *orthogonal* with respect to a given inner product $(\cdot, \cdot)$ if $(a, b) = 0$.

Example: In $\mathbb{R}^3$ $u = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$ $v = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$
These are orthogonal $(u, v) = (1)(-2) + (2)(1) = 0$.



5.3 Sequence of monic polynomials

Example 5.8

Take the space of polynomials of degree 2 or less and the IP

$$(f, g) = \int_{-1}^1 f(x)g(x)dx.$$

Let $p(x) \equiv 1$, $q(x) \equiv x$, $r(x) \equiv x^2 - 1/3$, and $f(x) = 3x - 4$

We can check that $(r, p) = 0$, and $(r, q) = 0$. We can then verify that $(r, f) = 0$. **Details:**

$$(r, p) = \int_{-1}^1 (x^2 - \frac{1}{3}) dx = \left[\frac{1}{3}x^3 - \frac{1}{3}x \right]_{-1}^1 = \frac{1}{3} - \frac{1}{3} + \frac{1}{3} - \frac{1}{3} = 0$$

$$(r, q) = \int_{-1}^1 x^3 - \frac{1}{3}x dx = \left[\frac{1}{4}x^4 - \frac{1}{6}x^2 \right]_{-1}^1 = \frac{1}{4} - \frac{1}{6} - \frac{1}{4} + \frac{1}{6} = 0$$

$$\text{Also } f(x) = 3q(x) - 4p(x). \text{ So } (r, f) = 3(r, q) - 4(r, p) = 0$$

5.3 Sequence of monic polynomials

As given above, a polynomial is **monic** if the coefficient of the leading term is 1:

$$p_n = x^n + c_{n-1}x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1x + c_0.$$

We'll now look at a sequence of such polynomials

$$\{\tilde{p}_0, \tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_n, \dots\}$$

that are orthogonal to each other:

$$(\tilde{p}_i, \tilde{p}_j) := \int_a^b \tilde{p}_i(x) \tilde{p}_j(x) dx = 0 \quad \text{if } i \neq j.$$

5.3 Sequence of monic polynomials

We want to establish some important facts about monic polys:

- ▶ A set of monic polynomials of degrees $1, 2, \dots, n$, forms a basis for $\mathcal{P}_n$.
- ▶ If the members of that set are orthogonal to each other, then they are orthogonal to *all* polynomials of lower degree.
- ▶ We can construct such as set.

5.3 Sequence of monic polynomials

Theorem 5.9

Let $\{\tilde{p}_i\}_{i=0}^n$ be a sequence of polynomials where each $\tilde{p}_i$ is monic and exactly of degree i . This sequence forms a basis for $\mathcal{P}_n$.

Proof: Recall that $\mathcal{P}_n$ is the space of all polynomials of degree n or less.

Also $\tilde{p}_0(x) \equiv 1$. If $n=0$, then any element of $\mathcal{P}_0$ is just a constant — and so a multiple of $\tilde{p}_0$. So $\tilde{p}_0$ is a basis for $\mathcal{P}_0$.

5.3 Sequence of monic polynomials

See bound.

5.3 Sequence of monic polynomials

Theorem 5.9 means that if q is a polynomial of degree n then it can be written uniquely as a linear combination of the $\tilde{p}_i$:

$$q(x) = \sum_{i=0}^n a_i \tilde{p}_i(x),$$

for some unique choice of the real coefficients a_i .

5.3 Sequence of monic polynomials

Definition 5.10 (Legendre Polynomials)

The sequence $\{\tilde{p}_i\}_{i=0}^n$ is a sequence of *monic, orthogonal* polynomials if each $\tilde{p}_i$ is monic and *exactly* of degree i and

$$(\tilde{p}_i, \tilde{p}_j) = 0 \quad \text{if } i \neq j.$$

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5.3 Sequence of monic polynomials

Theorem 5.11

If $\tilde{p}_j \in \{\tilde{p}_i\}_{i=0}^{\infty}$ then $\tilde{p}_j$ is orthogonal to all polynomials of degree less than j .

Proof: Let q be any polynomial of degree $j-1$ (or less).

Since $\{\tilde{p}_0, \tilde{p}_1, \dots, \tilde{p}_{j-1}\}$ is a basis for $\mathbb{P}_{j-1}$, we can write

$$q(x) = c_0 \tilde{p}_0 + c_1 \tilde{p}_1 + \dots + c_{j-1} \tilde{p}_{j-1}$$

Then

$$(q, \tilde{p}_j) = c_0 (\tilde{p}_0, \tilde{p}_j) + c_1 (\tilde{p}_1, \tilde{p}_j) + \dots + c_{j-1} (\tilde{p}_{j-1}, \tilde{p}_j)$$
$$= c_0(0) + c_1(0) + \dots + c_{j-1}(0) = 0$$

5.4 Constructing the Sequence

Theorem 5.12

*The sequence $\{\tilde{p}_i\}_{i=0}^{\infty}$ exists and can be constructed as follows:
Let α and β be defined as*

$$\alpha_{n+1} = \frac{(x\tilde{p}_n, \tilde{p}_n)}{(\tilde{p}_n, \tilde{p}_n)}, \quad \text{and} \quad \beta_{n+1} = \frac{(x\tilde{p}_n, \tilde{p}_{n-1})}{(\tilde{p}_{n-1}, \tilde{p}_{n-1})},$$

then the sequence is given by

$$\tilde{p}_0(x) \equiv 1, \quad \tilde{p}_1(x) = x - \alpha_1$$

and

$$\tilde{p}_{n+1}(x) = (x - \alpha_{n+1})\tilde{p}_n(x) - \beta_{n+1}\tilde{p}_{n-1}(x),$$

for $n \geq 1$.

The proof uses *Gram-Schmidt Orthogonalization*.

5.4 Constructing the Sequence

Idea: we want $\hat{p}_{n+1}$ to be orthogonal to $\hat{p}_0, \hat{p}_1, \dots$, and, specifically, $\hat{p}_{n-1}$ & $\hat{p}_n$.

Use $(\hat{p}_{n+1}, \hat{p}_n) = 0$. Therefore

$$\left((x - \alpha_{n+1}) \hat{p}_n + \beta_{n+1} \hat{p}_{n-1}, \hat{p}_n \right) = 0$$

$$\Rightarrow (x \hat{p}_n, \hat{p}_n) - \alpha_{n+1} (\hat{p}_n, \hat{p}_n) + \beta_{n+1} \underbrace{(\hat{p}_{n-1}, \hat{p}_n)}_0 = 0$$

$$\Rightarrow \alpha_{n+1} = \frac{(x \hat{p}_n, \hat{p}_n)}{(\hat{p}_n, \hat{p}_n)} \quad \text{as required.}$$

Now use that $(\hat{p}_{n+1}, \hat{p}_{n-1}) = 0$ to get β_{n+1} .



5.4 Constructing the Sequence

Example 5.13

If we use the inner product $(f, g) := \int_{-1}^1 f(x)g(x) dx$ then the first 3 polynomials in the sequence are:

$$\tilde{p}_0 = 1, \quad \tilde{p}_1 = x, \quad \text{and} \quad \tilde{p}_2 = x^2 - 1/3.$$

Check

$$\begin{aligned} (\tilde{p}_2, \tilde{p}_1) &= \int_{-1}^1 (x^2 - \frac{1}{3})x \, dx \\ &= \int_{-1}^1 x^3 - \frac{1}{3}x \, dx = \left(\frac{1}{4}x^4 - \frac{1}{6}x^2 \right) \Big|_{-1}^1 \\ &= \frac{1}{4} - \frac{1}{4} - \frac{1}{6} + \frac{1}{6} = 0. \quad \checkmark \end{aligned}$$

5.4 Constructing the Sequence

Example 5.14

Find the zeros of $\tilde{p}_2$ from Example 5.13.

$$\hat{p}_2 = x^2 - \frac{1}{3}.$$

$$\text{So } \hat{p}_2(x_i) = 0 \Rightarrow x_i^2 = \frac{1}{3} \Rightarrow x_i = \pm \frac{1}{\sqrt{3}}$$

So zeros are

$$x_0 = -\frac{1}{\sqrt{3}} \quad x_1 = \frac{1}{\sqrt{3}}$$

5.5 Properties of the sequence

One of the ways of constructing the Gaussian Quadrature rule $G_n(\cdot)$ on $n + 1$ is to take the quadrature points as the roots of $\tilde{p}_{n+1}$.

We know (from the fundamental theorem of algebra) a polynomial of degree $n + 1$ has exactly $n + 1$ roots in $\mathbb{C}$ up to multiplicity.

However, for the roots of $\tilde{p}_{n+1}$ to be feasible as quadrature points for estimating $\int_a^b f(x)dx$, they would need special properties:

- we need the roots, $x_0, x_1, \dots, x_n$ to be distinct.
- All must be real-valued.
- Need each $x_i \in [a, b]$

5.5 Properties of the sequence

Theorem 5.15

Let $\tilde{p}_i \in \{\tilde{p}_i\}_{i=0}^{\infty} = \{\tilde{p}_0, \tilde{p}_1, \dots\}$ be the set of monic polynomials that are orthogonal with respect to the inner product

$(u, v) := \int_a^b u(x)v(x)dx$. Then:

- (i) The zeros of each $\tilde{p}_i \in \{\tilde{p}_i\}_{i=0}^{\infty}$ are simple (not repeated).
- (ii) All the zeros of $\tilde{p}_i$ are real numbers in the interval $[a, b]$.

(i) Suppose that $\tilde{p}_i$ has a repeated root, c .
That is $\tilde{p}_i(x) = (x-c)^2 q(x)$ where $q(x)$ has degree $i-2$. Since $\deg(\tilde{p}_i) > \deg(q)$ we should have $(\tilde{p}_i, q) = 0$.
But $((x-c)^2 q, q) = \int_a^b (x-c)^2 q(x) q(x) dx$

5.5 Properties of the sequence

$$= \int_a^b \left[(x-c) q(x) \right]^2 dx > 0$$

Since $(x-c)(q(x))^2 \geq 0$ for all x .

Contradiction!

(ii) Next we show all the zeros are in $[a, b]$. Suppose $\hat{q}_i$ has k zeros in $[a, b]$ but $k < i$, so we can write

$$\hat{p}_i(x) = q(x) r(x) \quad \text{where } q(x)$$

has k zeros in $[a, b]$ and $r(x)$ has none.

So $r(x)$ has degree $< i$ and $r(x)$ does not change sign on $[a, b]$.

5.5 Properties of the sequence

So $(\hat{p}_i, q) = 0$. since $\deg(q) = k < i$

$$\begin{aligned}\text{But } (\hat{p}_i, q) &= \int_a^b q(x) r(x) q(x) dx \\ &= \int_a^b (q(x))^2 r(x) dx > 0\end{aligned}$$

Since $q(x)^2 r(x)$ does not change sign on $[a, b]$. 

5.6 Legendre Polynomials

The polynomials we've looked at are an example of **Legendre Polynomials**.

As we'll as been useful for numerical integration methods, they have many other applications, including in astronomy, electrical engineering, number theory, and neural networks.

5.6 Legendre Polynomials

FYI: The monic polynomials we studied in Chapter 3 are also known as Legendre Polynomials, named after Adrien-Marie Legendre.

This is the only known “portrait” of him, which was (re)discovered in 2008.



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5.7 Exercises

Exercise 5.1

$\mathcal{P}_n$, the space of polynomials of degree (at most) n forms a vector space. Is it true that the space of *monic* polynomials of degree n forms a vector space?

Exercise 5.2

Given a set of $n + 1$ distinct interpolation points, $\{x_0, x_1, \dots, x_n\}$, let $\{L_i\}_{i=0}^n$ be the set of associated Lagrange basis functions.

1. Show that the set $\{L_i\}_{i=0}^n$ is linearly independent.
2. What space do they span?
3. Give two other difference bases for the same space.

Exercise 5.3

Let $\tilde{p}_n$ be the n th Legendre Polynomial. Show that, if $n > 0$, then $\int_a^b \tilde{p}_n(x) dx = 0$.

5.7 Exercises

Exercise 5.4

(i) Using the Inner Product

$$(f, g) := \int_0^1 f(x)g(x)dx,$$

find $\tilde{p}_0(x)$, $\tilde{p}_1(x)$, $\tilde{p}_2(x)$ and $\tilde{p}_3(x)$.

(ii) Find the zeros of $\tilde{p}_2(x)$ and call them x_0 and x_1 . Construct a quadrature rule for $\int_0^1 f(x)dx$ taking these as the quadrature points, and the weights as the integrals to the corresponding Lagrange polynomials.